**ELEC70022**
**EE9-CS6-1**

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE PART IV: MEng and ACGI

# MODELLING AND CONTROL OF MULTI-BODY MECHANICAL SYSTEMS

Friday, 3 May 10:00 am

Time allowed: 3:00 hours

**There are FOUR questions on this paper.**

**Answer ALL questions**

*All questions carry equal marks.*

*This is an OPEN BOOK examination.*

**Any special instructions for invigilators and information for candidates are on page 1.**

Examiners responsible First Marker(s) : S.A. Evangelou
Second Marker(s) : D. Angeli

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# MODELLING AND CONTROL OF MULTIBODY MECHANICAL SYSTEMS

1. As shown in Figure 1.1, in spherical coordinates two angles $\theta$ and $\phi$ and a radial distance $r$ are used to specify the position of a point $A$. Two Cartesian coordinate systems are involved: a fixed Cartesian coordinate system with unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$, and $\boldsymbol{k}$, and a moving spherical coordinate system with unit vectors $\boldsymbol{e_r}$, $\boldsymbol{e_\theta}$, and $\boldsymbol{e_\phi}$.

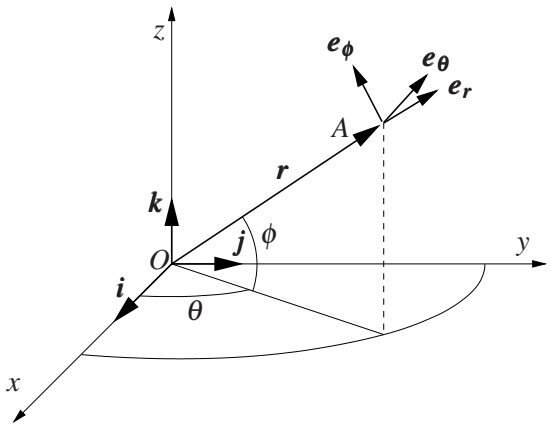



Figure 1.1 Position vector in spherical coordinates.

The spherical unit vectors written in fixed Cartesian coordinates are (from the notes)

$$\begin{aligned} \boldsymbol{e_r} &= \cos\theta\cos\phi\,\boldsymbol{i} + \sin\theta\cos\phi\,\boldsymbol{j} + \sin\phi\,\boldsymbol{k}, \\ \boldsymbol{e_\theta} &= -\sin\theta\,\boldsymbol{i} + \cos\theta\,\boldsymbol{j}, \\ \boldsymbol{e_\phi} &= -\cos\theta\sin\phi\,\boldsymbol{i} - \sin\theta\sin\phi\,\boldsymbol{j} + \cos\phi\,\boldsymbol{k}. \end{aligned} \tag{1.1}$$

The derivatives of the spherical unit vectors are given by (from the notes)

$$\begin{aligned} \dot{\boldsymbol{e}}_{\boldsymbol{r}} &= \dot\theta\cos\phi\,\boldsymbol{e_\theta} + \dot\phi\,\boldsymbol{e_\phi}, \\ \dot{\boldsymbol{e}}_{\boldsymbol{\theta}} &= -\dot\theta\cos\phi\,\boldsymbol{e_r} + \dot\theta\sin\phi\,\boldsymbol{e_\phi}, \\ \dot{\boldsymbol{e}}_{\boldsymbol{\phi}} &= -\dot\phi\,\boldsymbol{e_r} - \dot\theta\sin\phi\,\boldsymbol{e_\theta}. \end{aligned} \tag{1.2}$$

a) By choosing an appropriate Euler angle convention and using standard single-axis-rotation transformation matrices, calculate the transformation matrices $\boldsymbol{A_1}$ from fixed to spherical coordinates, and $\boldsymbol{A_2}$ from spherical to fixed coordinates. [ 8 ]

b) By using a transformation matrix calculated in part a), derive the same expressions as those in (1.1) for the spherical unit vectors in terms of the fixed unit vectors. [ 4 ]

c) Compute the angular velocity vector of the spherical coordinate system in the spherical coordinate system. [ 4 ]

d) By using the angular velocity vector computed in part c), calculate the derivatives of the spherical unit vectors and compare these derivatives with the expressions in (1.2). [ 4 ]

2. As shown in Figure 2.1, a heavy load of mass $m$ is suspended from a crane by massless and inextensible left and right wire ropes of length $l_r$ each, attached to the crane at points $A$ and $E$, respectively, where in the present exercise the crane (not shown) is assumed to be stationary (fixed to Earth). The heavy load is attached to the left and right wire ropes at points $B$ and $D$, respectively, with the distance between points $B$ and $D$ and between points $A$ and $E$ being the same and equal to $l_b$. The heavy load is free to swing and rotate in space under the influence of gravity and the left and right wire rope forces, each along the corresponding wire rope and assumed to always remain in tension. It is also assumed that the motion of the heavy load is not large enough to cause entanglement of the wire ropes with each other or of the heavy load with any or both of the wire ropes. Due to the relatively flat shape of the heavy load, its centre of mass, indicated by point $C$ in Figure 2.1, is assumed to be at its top surface in the middle of the distance between points $B$ and $D$.

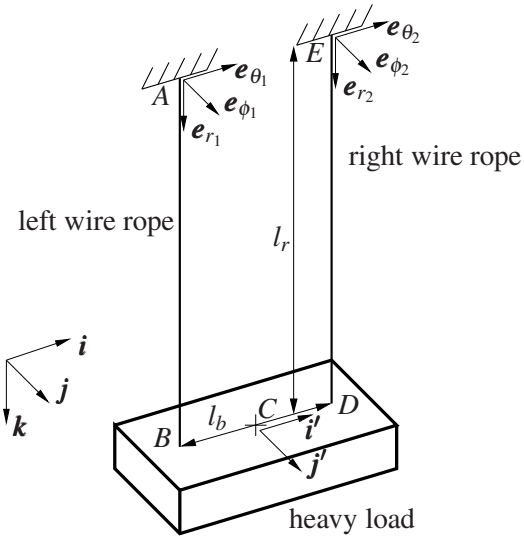



Figure 2.1 Heavy load suspended from a crane.

An Earth-fixed and three moving Cartesian coordinate systems are used to analyse the motion of the heavy load, as follows: a) an Earth-fixed Cartesian coordinate system with unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ and $\boldsymbol{k}$, with $\boldsymbol{i}$ horizontal and aligned with the line joining points $A$ and $E$, b) two spherical coordinate systems with unit vectors $\boldsymbol{e}_{r_1}$, $\boldsymbol{e}_{\theta_1}$, and $\boldsymbol{e}_{\phi_1}$ for the first, and $\boldsymbol{e}_{r_2}$, $\boldsymbol{e}_{\theta_2}$, and $\boldsymbol{e}_{\phi_2}$ for the second, attached respectively to the left and right wire ropes at points $A$ and $E$, and c) a heavy-load-fixed Cartesian coordinate system with its origin at point $C$ and its $\boldsymbol{i}'$, $\boldsymbol{j}'$, and $\boldsymbol{k}'$ (not shown) unit vectors initially respectively aligned with the Earth-fixed unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ and $\boldsymbol{k}$. The respective heavy-load-fixed axes correspond to principal axes with principal moments of inertia $I_{xx}$, $I_{yy}$, and $I_{zz}$. The angular configuration of the heavy load with respect to Earth is represented by three Euler angles $\psi$, $\theta$ and $\phi$ in the yaw-pitch-roll convention.

a) Write the generalised coordinates of the system. [ 2 ]

b) Write the position vector of the centre of mass of the heavy load from point $A$ in terms of the Earth-fixed unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ and $\boldsymbol{k}$. [ 4 ]

c) Calculate the velocity vector of the centre of mass of the heavy load in terms of the Earth-fixed unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ and $\boldsymbol{k}$. [ 2 ]

d) Compute the kinetic energy and potential energy of the heavy load, and hence

determine the Lagrangian function, $L$. [ 4 ]

e) Derive the kinematic constraints that restrict the motion of the heavy load. [ 4 ]

f) Write the Lagrangian equations of motion of the system (the Lagrangian function can be left as $L$ in the equations of motion, that is, there is no need to substitute it with the expression found for it in part d) and carry out any further calculations). [ 4 ]

3. An inverted uniform right circular hollow cone of radius $R$, height $h$, and mass $m$, is shown in Figure 3.1. The origin $O$ of the Cartesian coordinate system with axes $X$, $Y$, and $Z$ is at the vertex of the cone. The $Z$ axis is the axis of symmetry of the cone. The area of the curved surface of the cone (where all the mass is distributed) is $\pi RL$, in which $L$ is the slant height.

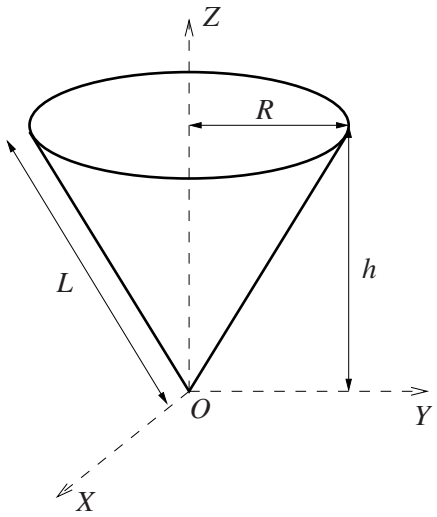



Figure 3.1 Right circular cone.

a) Write in cylindrical coordinates the infinitesimal area element of the area integration used to find the moment of inertia (Hint: think about the relationship between a coordinate along the slant height and the $z$ cylindrical coordinate). [ 5 ]

b) By using cylindrical coordinates, compute the moment of inertia of the cone in terms of $m$, $R$ and $h$ about the following axes:

i) the axis of symmetry of the cone; [ 7 ]

ii) the X axis; [ 6 ]

iii) the Y axis. [ 2 ]

4. As in Q2 and as shown in Figure 4.1, a heavy load of mass $m$ is suspended from a crane by massless and inextensible left and right wire ropes of length $l_r$ each, attached to the crane at points $A$ and $E$, respectively, where in the present exercise the crane (not shown) is assumed to be stationary (fixed to Earth). The heavy load is attached to the left and right wire ropes at points $B$ and $D$, respectively, with the distance between points $B$ and $D$ and between points $A$ and $E$ being the same and equal to $l_b$. The heavy load is free to swing and rotate in space under the influence of gravity and the left and right wire rope forces, each along the corresponding wire rope and assumed to always remain in tension. It is also assumed that the motion of the heavy load is not large enough to cause entanglement of the wire ropes with each other or of the heavy load with any or both of the wire ropes. Due to the relatively flat shape of the heavy load, its centre of mass, indicated by point $C$ in Figure 4.1, is assumed to be at its top surface in the middle of the distance between points $B$ and $D$.

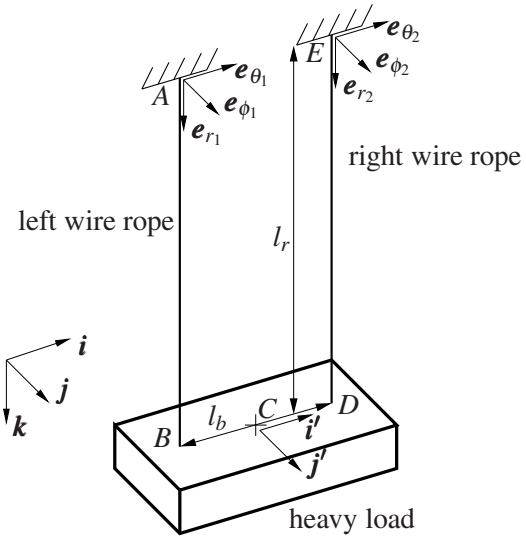



Figure 4.1 Heavy load suspended from a crane.

An Earth-fixed and three moving Cartesian coordinate systems are used to analyse the motion of the heavy load, as follows: a) an Earth-fixed Cartesian coordinate system with unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ and $\boldsymbol{k}$, with $\boldsymbol{i}$ horizontal and aligned with the line joining points $A$ and $E$, b) two spherical coordinate systems with unit vectors $\boldsymbol{e}_{r_1}$, $\boldsymbol{e}_{\theta_1}$, and $\boldsymbol{e}_{\phi_1}$ for the first, and $\boldsymbol{e}_{r_2}$, $\boldsymbol{e}_{\theta_2}$, and $\boldsymbol{e}_{\phi_2}$ for the second, attached respectively to the left and right wire ropes at points $A$ and $E$, and c) a heavy-load-fixed Cartesian coordinate system with its origin at point $C$ and its $\boldsymbol{i}'$, $\boldsymbol{j}'$, and $\boldsymbol{k}'$ (not shown) unit vectors initially respectively aligned with the Earth-fixed unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ and $\boldsymbol{k}$. The respective heavy-load-fixed axes correspond to principal axes with principal moments of inertia $I_{xx}$, $I_{yy}$, and $I_{zz}$. The angular configuration of the heavy load with respect to Earth is represented by three Euler angles $\psi$, $\theta$ and $\phi$ in the yaw-pitch-roll convention.

a) For a general unit vector $\boldsymbol{b}$ that rotates by an angular velocity vector $\boldsymbol{\Omega}_{\boldsymbol{b}}$, calculate its first and second derivatives with respect to time, $\dot{\boldsymbol{b}}$ and $\ddot{\boldsymbol{b}}$, respectively. [ 2 ]

b) Write the heavy load angular velocity vector with respect to Earth, $\boldsymbol{\Omega}$, in terms of the unit vectors $\boldsymbol{i}'$, $\boldsymbol{j}'$, and $\boldsymbol{k}'$ and the Euler angles $\psi$, $\theta$ and $\phi$, and then compute the heavy load angular acceleration vector with respect to Earth, $\dot{\boldsymbol{\Omega}}$,

in the same coordinate system. [ 3 ]

c) Write the relative position vector of the centre of mass, $C$, of the heavy load from the lower end of the left wire rope at point $B$, that is, write the position vector $\boldsymbol{r}_{BC}$, in the heavy-load-fixed coordinate system. Then calculate the relative velocity and relative acceleration vectors, $\dot{\boldsymbol{r}}_{BC}$ and $\ddot{\boldsymbol{r}}_{BC}$, respectively, in the same coordinate system. [ 3 ]

d) By assuming $\ddot{\boldsymbol{r}}_{BC}$ in part c) is calculated, explain how the acceleration vector of the heavy load centre of mass with respect to Earth can be computed in Earth-fixed coordinates. [ 3 ]

e) By considering the kinematic relationship between points $A$, $B$, $D$, and $E$, derive the equations of constraint that the wire ropes impose on the heavy load. [ 3 ]

f) Assume the acceleration vector of the heavy load centre of mass has been calculated as $a_x\boldsymbol{i} + a_y\boldsymbol{j} + a_z\boldsymbol{k}$, where $a_x$, $a_y$, and $a_z$ are functions of the generalised coordinates. By considering the translational motion of the centre of mass of the heavy load, use the vectorial approach to derive equations relating any unknown constraint forces acting on the heavy load and the generalised coordinates. [ 3 ]

g) Describe the remaining steps required to obtain the motion of the heavy load and also write the number of degrees of freedom of the system. [ 3 ]